

# Simplifying expressions with rational exponents

The index laws applied to letters — and the factorising tricks that work just as well with fractional powers.

LESSON 32–33

2 × 40 MIN

ALGEBRA 8, §10

STAGE 9 · 2.3

## LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up 5 min

Explanation 12 min

Interactive 8 min

Practice 13 min

Homework 2 min

## LEARNING OBJECTIVES

- Simplify products and quotients of powers with rational exponents.
- Take a common power out as a factor.
- Recognise a difference of squares in  $a - b = (\sqrt{a})^2 - (\sqrt{b})^2$ .
- Write an answer in root form when asked.

## TEXTBOOK

National §10, pp. 49–52

Cambridge Learner's Book pp. 30–33

Workbook Workbook 2.3

## 01 Explanation

### The routine

1. Turn every root into a power. Fractions of indices are easier than nested radicals.
2. Apply the index laws.
3. Collect, factorise or cancel.
4. Convert back to root form only if the question asks for it.

$$\frac{\sqrt{a} \cdot \sqrt[3]{a}}{\sqrt[6]{a}} = \frac{a^{1/2} \cdot a^{1/3}}{a^{1/6}} = a^{1/2 + 1/3 - 1/6} = a^{2/3}$$

### Taking out a common power

The smallest index is the one to take out — exactly as with whole-number powers:

$$a^{3/2} + a^{1/2} = a^{1/2}(a + 1)$$

Check by multiplying back:  $a^{1/2} \cdot a = a^{3/2}$  ✓ and  $a^{1/2} \cdot 1 = a^{1/2}$  ✓.

### The difference-of-squares trick

Since  $a = (\sqrt{a})^2$  for  $a \geq 0$ , any difference of two non-negative numbers is a difference of squares:

$$a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$$

That is the single most useful move in this section. It cancels fractions such as

$$\frac{a - b}{\sqrt{a} - \sqrt{b}} = \sqrt{a} + \sqrt{b}$$

and it also rationalises a two-term denominator, by multiplying by the **conjugate**:

$$\frac{1}{\sqrt{5} - \sqrt{3}} = \frac{\sqrt{5} + \sqrt{3}}{(\sqrt{5})^2 - (\sqrt{3})^2} = \frac{\sqrt{5} + \sqrt{3}}{2}$$

## 02 Worked examples

**EXAMPLE 1** Simplify  $\frac{a^{3/4} \cdot a^{1/2}}{a^{1/4}}$

①  $\frac{3}{4} + \frac{1}{2} = \frac{5}{4}$

Multiply on top: add the indices.

②  $\frac{5}{4} - \frac{1}{4} = 1$

Divide: subtract the index.

③  $= a$

ANSWER

$a$

**EXAMPLE 2** Factorise  $a^{3/2} - a^{1/2}$

① smallest index is  $\frac{1}{2}$

That is the common factor.

②  $a^{1/2}(a^1 - 1)$

Subtract  $\frac{1}{2}$  from each index.

③  $= a^{1/2}(a - 1)$   
or  $\sqrt{a}(a - 1)$

ANSWER

$\sqrt{a}(a - 1)$

**EXAMPLE 3** Simplify  $\frac{x-9}{\sqrt{x}-3}$ ,  $x \geq 0, x \neq 9$

①  $x = (\sqrt{x})^2$  and  $9 = 3^2$

A difference of squares in disguise.

②  $x - 9 = (\sqrt{x} - 3)(\sqrt{x} + 3)$

③  $= \sqrt{x} + 3$

Cancel the bracket.

ANSWER

$$\sqrt{x} + 3$$

### 03 Interactive model

Use the model to check any answer numerically: put a value in and compare both forms.

Check an index answer numerically

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value<sup>3</sup> **64**

a<sup>1</sup> **64**

Raising the answer to the power 3 gives back a<sup>1</sup> — that is exactly what “root of degree 3” means.

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH               | O'ZBEKCHA           | РУССКИЙ            |
|-----------------------|---------------------|--------------------|
| Simplify              | Soddalashtirish     | Упростить          |
| Common power          | Umumiy daraja       | Общая степень      |
| Substitution          | Almashtirish        | Замена             |
| Difference of squares | Kvadratlar ayirmasi | Разность квадратов |
| Factor out            | Qavsdan chiqarish   | Вынести за скобку  |
| Root form             | Ildiz ko'rinishi    | Форма с корнем     |
| Index form            | Daraja ko'rinishi   | Степенная форма    |
| Expression            | Ifoda               | Выражение          |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

**Quick check**
**Check yourself**
 $a^{1/2} \cdot a^{1/2} \cdot a^{1/2}$  equals:

$a^{1/8}$

$a^{3/2}$

$3a^{1/2}$

$a^{1/6}$

 $a^{3/2} + a^{1/2}$  factorises to:

$a^{1/2}(a + 1)$

$a(a^{1/2} + 1)$

$a^2(a + 1)$

it does not factorise

 $\frac{x-4}{\sqrt{x}} - 2$  equals:

$\sqrt{x} - 2$

$\sqrt{x} + 2$

$x + 2$

$2$

To rationalise  $\frac{1}{\sqrt{5} - \sqrt{3}}$  multiply by:

$$\sqrt{5}$$

$$\sqrt{5} + \sqrt{3}$$

$$\sqrt{5} - \sqrt{3}$$

$$2$$

## 06 Practice · 21 problems

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Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify  $a^{1/2} \cdot a^{1/2}$

ANSWER  $a$

2. Simplify  $a^{3/4} : a^{1/4}$

ANSWER  $a^{1/2}$

3. Simplify  $(a^{1/3})^3$

ANSWER  $a$

4. Simplify  $a^{1/2} \cdot a$

ANSWER  $a^{3/2}$

5. Write  $\sqrt{a} \cdot \sqrt{a}$  more simply

ANSWER  $a$

6. Simplify  $(4a^2)^{1/2}$ ,  $a \geq 0$

ANSWER  $2a$

7. Simplify  $a^{1/2} + 3a^{1/2}$

ANSWER  $4a^{1/2}$

Medium · the standard question

7 PROBLEMS



1. Simplify  $\frac{a^{3/4} \cdot a^{1/2}}{a^{1/4}}$

ANSWER  $a$

2. Factorise  $a^{3/2} - a^{1/2}$

ANSWER  $\sqrt{a}(a - 1)$

3. Simplify  $\frac{x - 9}{\sqrt{x} - 3}$

ANSWER  $\sqrt{x} + 3$

4. Simplify  $(a^{1/2}b^{1/2})^4$

ANSWER  $a^2b^2$

5. Simplify  $\sqrt{a} \cdot \sqrt[3]{a}$

ANSWER  $a^{5/6}$

6. Simplify  $\frac{a}{\sqrt{a}}$ ,  $a > 0$

ANSWER  $\sqrt{a}$

7. Factorise  $a + a^{1/2}$

ANSWER  $a^{1/2}(a^{1/2} + 1)$

Hard · stretch and reason

7 PROBLEMS

1. Simplify  $\frac{\sqrt{a} \cdot \sqrt{3a}}{\sqrt{6a}}$

ANSWER  $a^{2/3}$

2. Simplify  $\frac{x-1}{\sqrt{x}-1}$ ,  $x \geq 0, x \neq 1$

ANSWER  $\sqrt{x} + 1$

3. Rationalise  $\frac{1}{\sqrt{5}-\sqrt{3}}$

ANSWER  $\frac{\sqrt{5} + \sqrt{3}}{2}$

4. Simplify  $(a^{2/3} - b^{2/3}) : (a^{1/3} - b^{1/3})$

ANSWER  $a^{1/3} + b^{1/3}$

5. Simplify  $\frac{a^{3/2} - a^{1/2}}{a - 1}$ ,  $a > 0, a \neq 1$

ANSWER  $a^{1/2}$

6. Evaluate  $(\sqrt{7} + \sqrt{3})(\sqrt{7} - \sqrt{3})$

ANSWER  $7 - 3 = 4$

7. Simplify  $(a^{1/2} + 1)^2 - (a^{1/2} - 1)^2$

ANSWER  $4a^{1/2}$

## 07 Homework

### Homework — 5 problems

Algebra 8, §10, pp. 49–52.

1. Simplify  $a^{2/3} \cdot a^{1/3}$

2. Simplify  $\frac{b^{7/4}}{b^{3/4}}$

3. Factorise  $x^{3/2} + x^{1/2}$

4. Simplify  $\frac{x - 16}{\sqrt{x} - 4}$

5. Rationalise  $\frac{2}{\sqrt{7} - \sqrt{5}}$